

Søren Riis

Academic Curriculum Vitae

Reader in Computer Science, Queen Mary University of London

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Research profile

Mathematician and computer scientist working on mathematical logic, proof complexity, information theory, combinatorics and the foundations of learning. Recent work connects the limits of reinforcement learning, expert-level AI evaluation, original term-coding research and interactive presentations of theory with accompanying Lean-checked proofs.

Appointments and research visits

2000-present | Queen Mary University of London. Current role: Reader in Computer Science, School of EECS.

Jun 2018-Mar 2019 | Fetch.AI, Cambridge. Part-time Research Scientist Consultant; consensus protocols, hard-fork models, governance and distributed-ledger mechanisms.

1998-2000 | BRICS International PhD School, Aarhus University. Associate Professor; research and doctoral-level teaching.

1998 | The Fields Institute, Toronto. Research Fellow.

1995-1997 | Carlsberg-supported visiting research fellowship. Research visits and lecture series in Lund, Copenhagen, Ohio, Jerusalem and Leeds.

1994-1995 | BRICS International PhD School, Aarhus University. Research Fellow; research and doctoral-level teaching.

Additional research fellowships and visits include the Isaac Newton Institute, Cambridge, and Institut Henri Poincaré, Paris.

Education

University of Oxford | DPhil in Mathematics, Mathematical Logic, 1994

University of Copenhagen | MSc in Mathematics and Physics, 1989

Award

Humanity's Last Exam prize, US\$5,000 (2025). Contributed expert questions to the benchmark; member of the HLE Contributors Consortium on the Nature paper published in 2026.

Research leadership and funding

- Co-founder and director of the Centre for Discrete Mathematics at Queen Mary University of London.
- Research leadership on EPSRC grant EP/H016015/1, on information flows and bottlenecks, awarded £414,367; support for research staff and doctoral research.

Selected research contributions

Logic and proof complexity. Classified implications between the modular counting principles studied in $\text{Count}(q)$ Does Not Imply $\text{Count}(p)$; developed complexity-gap results for tree resolution and work on resolution with Stefan Dantchev.

Information theory and network coding. Developed graph guessing-game methods linking local information constraints to global information flow; collaborative work on guessing graphs, Boolean-network fixed points and non-Shannon information inequalities.

Combinatorics and collective choice. With Charles Leedham-Green and Klas Markström, proved that the largest Condorcet domain on eight alternatives has 224 orders and is unique up to isomorphism. Related work covers classification, heuristic search and research software.

AI evaluation and reinforcement learning. With Bei Zhou, studied AlphaZero-style learning in impartial games and distinguished champion performance from expert mastery. Further work explores multi-frame learning and mathematical models of drift and selection in LLM text ecosystems.

Interactive research and software

Interactive term-coding paper and Lean proof explorer | 2026

Original mathematical research on the boundary between polynomial-time computation and undecidability with disequality constraints. The interactive edition links the paper to explanations, formal statements, proof states and dependencies in an accompanying Lean-checked formalisation. External assumptions and the scope of verification are documented.

[Interactive paper](#) | [Lean source](#)

Gomulda AI search and learning lab | 2026

Exact graph-colouring optimisation, Monte Carlo tree search and minimax endgame search, with a separate experimental neural agent trained through self-play. Browser tools support training, checkpoints and comparisons. Based on the game invented by Cristoffer Crone.

[Live project](#) | [Source](#)

Solar Eclipse Explorer | 2026-present

Scientific computation and visualisation for the 2 August 2027 eclipse: local circumstances, totality duration, obscuration and the Sun's position, using NASA/GSFC Besselian elements.

[Live explorer](#) | [Source](#)

Further resources include the seven-alternative Condorcet atlas and interactive tools for guessing games, impartial games, waiting-time comparisons and text-ecosystem models.

[Research tools](#) | [Data and software](#)

Teaching and doctoral supervision

- Undergraduate and postgraduate teaching in logic, discrete structures, complexity and computation, with 3,000+ students taught.
- Module organisation for large cohorts and supervision and mentoring of student projects.
- Modules taught include ECS407U, Introduction to Logic and Discrete Structures, and ECS651U, Complexity and Computation.
- Doctoral supervision includes Stefan Dantchev, Sun Yun and Bei Zhou; involvement in the EPSRC Centre for Doctoral Training in Intelligent Games and Game Intelligence (IGGI).

Academic service

- EPSRC College of Peer Reviewers; member since 2010. Grant-review panel service in the UK and internationally.
- 50+ refereeing reports across logic, complexity, information theory, combinatorics and theoretical computer science.
- External examination and assessment of PhD research and student projects.

Selected talks and lectures

2026 | *Coin racing and waiting-time paradoxes: why fair coins are exceptional*. DIMAP Seminar, University of Warwick, 26 January.

2016 | *Finite Dynamical Systems, Causal Networks and Information Inequalities*. Lecture series, Institut Henri Poincaré, Paris, February.

2015 | *Ada Lovelace, a scientist in the archives*. Ada Lovelace 200 Symposium, University of Oxford, 9 December; with Ursula Martin and an introductory contribution by Nick Woodhouse.

2013 | *Graph guessing games and non-Shannon information inequalities*. Workshop on Entropy and Information Inequalities, Chinese University of Hong Kong, 17 April.

2013 | *Network Coding with Distributed Coding Functions*. Chinese University of Hong Kong EECS Seminar, 12 April.

2012 | *Graph Entropy, Network Coding and Guessing Games*. ZiF Workshop, Bielefeld, November.

2012 | *Valiant's Shift Problem*. Isaac Newton Institute, Cambridge, 9 May.

2012 | *A logician's approach to Network Coding*. Isaac Newton Institute, Cambridge, 10 February.

1998 | *Complexity Gaps for Unsatisfiability Problems*. BRICS Seminar, Aarhus, 14 December.

1998 | *Complexity Gaps for Nullstellensatz Proofs*. Fields Institute Workshop, Toronto.

1996 | *More on the relative strength of counting principles*. DIMACS Workshop, Rutgers University, May.

[Available recordings and slides](#)

Publications

Journal articles

1. Center for AI Safety, Scale AI and HLE Contributors Consortium (including S. Riis). A benchmark of expert-level academic questions to assess AI capabilities. *Nature* 649, 1139-1146 (2026). [Read](#)
2. B. Zhou, **S. Riis**. Impartial Games: A Challenge for Reinforcement Learning. *Machine Learning* 115, article 54 (2026). [Read](#)
3. B. Zhou, **S. Riis**. An efficient heuristic search algorithm for discovering large Condorcet domains. *4OR* 23, 193-216 (2025). [Read](#)
4. A. Karpov, K. Markström, **S. Riis**, B. Zhou. Coherent domains and improved lower bounds for the maximum size of Condorcet domains. *Discrete Applied Mathematics* 370, 57-70 (2025). [Read](#)
5. D. Akello-Egwel, C. R. Leedham-Green, A. Litterick, K. Markström, **S. Riis**. Condorcet domains on at most seven alternatives. *Mathematical Social Sciences* 133, 23-33 (2025). [Read](#)
6. B. Zhou, K. Markström, **S. Riis**. CDL: A fast and flexible library for the study of permutation sets with structural restrictions. *SoftwareX* 28, article 101951 (2024). [Read](#)
7. C. R. Leedham-Green, K. Markström, **S. Riis**. The largest Condorcet domain on 8 alternatives. *Social Choice and Welfare* 62, 109-116 (2024). [Read](#)
8. **S. Riis**, M. Gadouleau. Max-flow min-cut theorems on dispersion and entropy measures for communication networks. *Information and Computation* 267, 49-73 (2019). [Read](#)
9. R. Baber, D. Christofides, N. A. Dang, E. R. Vaughan, **S. Riis**. Graph Guessing Games and Non-Shannon Information Inequalities. *IEEE Transactions on Information Theory* 63, 4257-4267 (2017). [Read](#)
10. P. J. Cameron, N. A. Dang, **S. Riis**. Guessing Games on Triangle-Free Graphs. *Electronic Journal of Combinatorics* 23, P1.48 (2016). [Read](#)
11. M. Gadouleau, A. Richard, **S. Riis**. Fixed Points of Boolean Networks, Guessing Graphs, and Coding Theory. *SIAM Journal on Discrete Mathematics* 29, 2312-2335 (2015). [Read](#)
12. M. Gadouleau, **S. Riis**. Memoryless computation: New results, constructions, and extensions. *Theoretical Computer Science* 562, 129-145 (2015). [Read](#)
13. **S. Riis**. What makes a chess program original? Revisiting the Rybka case. *Entertainment Computing* 5, 189-204 (2014). [Read](#)
14. P. J. Cameron, M. Gadouleau, **S. Riis**. Combinatorial representations. *Journal of Combinatorial Theory, Series A* 120, 671-682 (2013). [Read](#)
15. M. Gadouleau, **S. Riis**. Graph-Theoretical Constructions for Graph Entropy and Network Coding Based Communications. *IEEE Transactions on Information Theory* 57, 6703-6717 (2011). [Read](#)
16. T. Wu, P. J. Cameron, **S. Riis**. On the guessing number of shift graphs. *Journal of Discrete Algorithms* 7, 220-226 (2009). [Read](#)
17. **S. Riis**. Information Flows, Graphs and their Guessing Numbers. *Electronic Journal of Combinatorics* 14, R44 (2007). [Read](#)
18. **S. Riis**. Reversible and Irreversible Information Networks. *IEEE Transactions on Information Theory* 53, 4339-4349 (2007). [Read](#)
19. **S. Riis**. A complexity gap for tree resolution. *Computational Complexity* 10, 179-209 (2001). [Read](#)
20. **S. Riis**, M. Sitharam. Generating hard tautologies using predicate logic and the symmetric group. *Logic Journal of the IGPL* 8, 787-795 (2000). [Read](#)
21. **S. Riis**. Bootstrapping the primitive recursive functions by only 27 colors. *Discrete Mathematics* 169, 269-272 (1997). [Read](#)
22. **S. Riis**. Count(q) Does Not Imply Count(p). *Annals of Pure and Applied Logic* 90, 1-56 (1997). [Read](#)
23. **S. Riis**. Count(q) versus the pigeon-hole principle. *Archive for Mathematical Logic* 36, 157-188 (1997). [Read](#)

Conference papers chapters and abstracts

24. **S. Riis**, U. Martin, N. Woodhouse. Ada Lovelace, a scientist in the archives. *Ada Lovelace Symposium*, 14 (2015). [Read](#)
25. R. Baber, D. Christofides, N. A. Dang, **S. Riis**, E. R. Vaughan. Multiple unicasts, graph guessing games, and non-Shannon inequalities. *International Symposium on Network Coding*, 1-6 (2013). [Read](#)
26. **S. Riis**, M. Gadouleau. A dispersion theorem for communication networks based on term sets. *IEEE International Symposium on Information Theory*, 593-597 (2011). [Read](#)
27. M. Gadouleau, **S. Riis**. Max-flow min-cut theorem for Rényi entropy in communication networks. *IEEE International Symposium on Information Theory*, 603-607 (2011). [Read](#)
28. **S. Riis**. On the Asymptotic Nullstellensatz and Polynomial Calculus Proof Complexity. *IEEE Symposium on Logic in Computer Science*, 272-283 (2008). [Read](#)
29. **S. Riis**, R. Ahlswede. Problems in Network Coding and Error Correcting Codes Appended by a Draft Version of S. Riis "Utilising Public Information in Network Coding". *GTIT-C*, 861-897 (2006). [Read](#)
30. S. S. Dantchev, **S. Riis**. On Relativisation and Complexity Gap. *Computer Science Logic*, 142-154 (2003). [Read](#)
31. S. S. Dantchev, **S. Riis**. "Planar" Tautologies Hard for Resolution. *IEEE Symposium on Foundations of Computer Science*, 220-229 (2001). [Read](#)
32. S. S. Dantchev, **S. Riis**. Tree Resolution Proofs of the Weak Pigeon-Hole Principle. *IEEE Conference on Computational Complexity*, 69-75 (2001). [Read](#)
33. K. Hägele, C. Ó'Dúnlaing, **S. Riis**. The complexity of scheduling TV commercials. *Mathematical Foundations of Computer Science and Information Technology*, 162-185 (2000). [Read](#)
34. P. Beame, **S. Riis**. More on the relative strength of counting principles. *Proof Complexity and Feasible Arithmetics*, 13-35 (1996). [Read](#)
35. A. Andersson, P. B. Miltersen, **S. Riis**, M. Thorup. Static Dictionaries on AC^0 RAMs: Query Time $\Theta(\sqrt{\frac{\log n}{\log \log n}})$ is Necessary and Sufficient. *IEEE Symposium on Foundations of Computer Science*, 441-450 (1996). [Read](#)

Preprints

- 36. **S. Riis**. One Unit Separates Polynomial Time from Undecidability in Term Coding. Author manuscript, revised 12 September 2026. [Manuscript](#) | [Interactive edition](#)
- 37. **S. Riis**, M. Paterson. Coin flipping and waiting times paradoxes: Why fair coins are exceptional. Preprint, arXiv:2601.16580 (2026). [Read](#)
- 38. **S. Riis**. Drift and selection in LLM text ecosystems. Preprint, arXiv:2604.08554 (2026). [Read](#)
- 39. A. Karpov, K. Markström, **S. Riis**, B. Zhou. Improved lower bounds for the maximum size of Condorcet domains. Preprint, arXiv:2601.07336 (2026). [Read](#)
- 40. **S. Riis**. Mastering NIM and Impartial Games with Weak Neural Networks: An AlphaZero-inspired Multi-Frame Approach. Preprint, arXiv:2411.06403 (2026). First submitted 2024; updated 2026. [Read](#)
- 41. **S. Riis**. Term Coding: An Entropic Framework for Extremal Combinatorics and the Guessing-Number Sandwich Theorem. Preprint, arXiv:2601.16614 (2026). [Read](#)
- 42. K. Markström, **S. Riis**, B. Zhou. Arrow's single peaked domains, richness, and domains for plurality and the Borda count. Preprint, arXiv:2401.12547 (2024). [Read](#)
- 43. A. Karpov, K. Markström, **S. Riis**, B. Zhou. Local Diversity of Condorcet Domains. Preprint, arXiv:2401.11912 (2024). [Read](#)
- 44. B. Zhou, **S. Riis**. Exploring Parity Challenges in Reinforcement Learning through Curriculum Learning with Noisy Labels. Preprint, arXiv:2312.05379 (2023). [Read](#)
- 45. **S. Riis**. Graph Entropy, Network Coding and Guessing games. Preprint, arXiv:0711.4175 (2007). [Read](#)

Technical reports

- 46. **S. Riis**. On the asymptotic Nullstellensatz and Polynomial Calculus proof complexity. *Electronic Colloquium on Computational Complexity*, TR09-014 (2009). [Read](#)
- 47. **S. Riis**, M. Sitharam. Non-constant Degree Lower Bounds imply linear Degree Lower Bounds. *Electronic Colloquium on Computational Complexity*, TR97-048 (1997). [Read](#)

Editorial

- 48. C. Butz, U. Kohlenbach, **S. Riis**, G. Winskel. Preface. *Annals of Pure and Applied Logic* 104, 1-2 (2000). [Read](#)